

Probability Space Exercises

Mathematics for AI

Question 1. Drawing cards

A pack of cards contains 52 cards. Assume each card has a number from 1 to 13 and a suit of Hearts, Clubs, Diamonds or Spades.

(a)

We draw one card from the pack, replace it, and then draw another. What are the chances that both cards are Hearts? What are the chances that the cards are different suits? Express your answers using a probability function P .

$$\begin{aligned}S &= \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\} \\N &= \{1, 2, \dots, 13\} \\D &= N \times S \\H &= N \times \{\heartsuit\} \subseteq D \\\Omega &= D \times D \\|\Omega| &= 52 \cdot 52 = 2704\end{aligned}$$

Let $(c_1, c_2) \in \Omega$ where each $c_i \in D$ is a single card. For each card, $c_i = (n_i, s_i)$ where $n_i \in N$ is its number and $s_i \in S$ is its suit.

Both hearts:

There are 13 different heart cards. Let H_1 be the set of pairs where the first card is a heart and let H_2 be the set of pairs where the second card is a heart.

$$\begin{aligned}H_1 &= \{(c_1, c_2) \in \Omega : c_1 \in H\} \\H_2 &= \{(c_1, c_2) \in \Omega : c_2 \in H\}\end{aligned}$$

$$P(H_1) = P(H_2) = \frac{13}{52}$$

So we can use the fact that if we draw 2 hearts that H_1 and H_2 are independent due to the replacement.

$$P(H_1 \cap H_2) = P(H_1)P(H_2) = \frac{13}{52} \cdot \frac{13}{52} = \frac{169}{2704} = \frac{1}{16}$$

Different suits:

Let Z be the event where 2 cards of different suits are drawn one after the other:

$$Z = \{(c_1, c_2) \in \Omega : s_1 \neq s_2\}$$

$$\begin{aligned}P(\Omega \setminus Z) &= 1 - P(Z) \\P(Z) &= 1 - P(\Omega \setminus Z) \\&= 1 - \frac{52 \cdot 13}{2704} \\&= 1 - \frac{1}{4} = \frac{3}{4}\end{aligned}$$

Note the $52 \cdot 13$ above here we are using the product rule that for the first card we have 52 choices but for the second we only have 13 as it needs to come from the same suit as that first card.

Summary: The probability of drawing two hearts is $P(H_1 \cap H_2) = \frac{1}{16}$ and the probability of drawing two cards with different suits is $P(Z) = \frac{3}{4}$.

(b)

Now we draw one card from the pack, do not replace it, and then draw another. What are the chances that both cards are Hearts? What are the chances that the cards are different suits? Again, express your answers using a probability function P .

$$\begin{aligned} S &= \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\} \\ N &= \{1, 2, \dots, 13\} \\ D &= N \times S \\ H &= N \times \{\heartsuit\} \subseteq D \\ \Omega &= \{(c_1, c_2) \in D \times D : c_1 \neq c_2\} = (D \times D) \setminus \{(c, c) : c \in D\} \\ |\Omega| &= 52 \cdot 51 = 2652 \end{aligned}$$

Let $(c_1, c_2) \in \Omega$ where each $c_i \in D$ is a single card. For each card, $c_i = (n_i, s_i)$ where $n_i \in N$ is its number and $s_i \in S$ is its suit.

Both hearts:

Option 1:

There are 13 different heart cards. Let H_1 be the set of pairs where the first card is a heart and let H_2 be the set of pairs where the second card is a heart.

$$\begin{aligned} H_1 &= \{(c_1, c_2) \in \Omega : c_1 \in H\} \\ H_2 &= \{(c_1, c_2) \in \Omega : c_2 \in H\} \end{aligned}$$

$$P(H_1 \cap H_2) = P(H_1)P(H_2|H_1) = \frac{13}{52} \cdot \frac{12}{51} = \frac{156}{2652} = \frac{1}{17}$$

Option 2:

We can confirm this using the complement. I did find this to be trickier as then I realized that the complement of Q meant that at least one card is not a heart. So we had to consider 3 cases: 1. where both are not hearts, 2. where the first card is a heart and the second is not and 3. where the first card is not a heart and the second card is a heart. Those are all disjoint events so we add their probabilities together.

$$\begin{aligned} P(\Omega \setminus (H_1 \cap H_2)) &= 1 - P(H_1 \cap H_2) \\ P(H_1 \cap H_2) &= 1 - P(\Omega \setminus (H_1 \cap H_2)) \\ &= 1 - \frac{((13 \cdot 39) + (39 \cdot 13) + (39 \cdot 38))}{52 \cdot 51} \\ &= 1 - \frac{2496}{2652} \\ &= 1 - \frac{64}{68} = 1 - \frac{16}{17} = \frac{1}{17} \end{aligned}$$

Different suits:

Let Z be the event where 2 cards of different suits are drawn one after the other:

$$Z = \{(c_1, c_2) \in \Omega : s_1 \neq s_2\}$$

$$\begin{aligned} P(\Omega \setminus Z) &= 1 - P(Z) \\ P(Z) &= 1 - P(\Omega \setminus Z) \\ &= 1 - \frac{52 \cdot 12}{52 \cdot 51} \\ &= 1 - \frac{624}{2652} \\ &= 1 - \frac{4}{17} = \frac{13}{17} \end{aligned}$$

Note the $52 \cdot 12$ above here we are using the product rule that for the first card we have 52 choices but for the second we only have 12 as it needs to come from the same suit as that first card and there is no replacement.

Summary: The probability, without replacing the cards, of drawing two hearts is $P(H_1 \cap H_2) = \frac{1}{17}$ and the probability of drawing two cards with different suits is $P(Z) = \frac{13}{17}$.

Question 2. Bayes' Theorem: smoke and fire

Dangerous fires are uncommon (0.5%), smoke is fairly common (15%), and 90% of dangerous fires make smoke. Use Bayes' Theorem to determine the percentage of occasions on which smoke means a dangerous fire.

Events and given probabilities.

- Let D = Dangerous fire and $P(D) = 0.5\%$
- Let ND = Not a dangerous fire and $P(ND) = 99.5\%$
- Let S = Smoke and $P(S) = 15\%$
- Let NS = No Smoke and $P(NS) = 85\%$
- $P(S | D) = 90\%$

Bayes' Theorem.

$$P(A | B) = \frac{P(A) P(B | A)}{P(B)}$$

Working.

$$\begin{aligned} P(D | S) &= \frac{P(D) P(S | D)}{P(S)} \\ &= \frac{(0.005)(0.9)}{(0.15)} \\ &= \frac{0.0045}{0.15} \\ &= 0.03 = 3\% \end{aligned}$$

This means that there is only a 3% chance that if you see smoke that there is a dangerous fire. This makes sense as dangerous fires are much less common than smoke, so in 97% of cases when you see smoke it is coming from a harmless source.

Question 3. Proving three properties of a probability space

Recall the three stated properties of a probability space $(\Omega, \mathcal{F}, P)$:

1. $P(\Omega \setminus A) = 1 - P(A)$
2. If $A \subseteq B$ then $P(B) = P(A) + P(B \setminus A) \geq P(A)$
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Using the definition of a probability space and the set theory from Week 2, prove each property.

Property 1: $P(\Omega \setminus A) = 1 - P(A)$

Proof. Let $A \in \mathcal{F}$. From axiom 2 of the definition of the σ -field: $\Omega \setminus A \in \mathcal{F}$ and so $P(\Omega \setminus A)$ is defined.

If $x \in A \cap (\Omega \setminus A)$ then $x \in A$ and $x \in (\Omega \setminus A)$. From the definition of the complement of A in Ω this can be stated as $x \in A$ and $x \in \Omega \wedge x \notin A$. This is a contradiction: $x \in A \wedge x \notin A$. So $A \cap (\Omega \setminus A) = \emptyset$.

If $x \in A \cup (\Omega \setminus A)$ then $x \in A$ or $x \in (\Omega \setminus A)$. $A \subseteq \Omega$ as $\mathcal{F}$ is a subset of the power set of Ω , so if $x \in A$ then $x \in \Omega$. But if $x \in (\Omega \setminus A)$, from the definition of the complement, $x \in \Omega \wedge x \notin A$, which also gives $x \in \Omega$. Considering the converse, if $x \in \Omega$ then either $x \in A$, or $x \notin A$ and so $x \in \Omega \setminus A$. So from either case $x \in A \vee x \in \Omega \setminus A$, which from the definition of the union is $x \in A \cup (\Omega \setminus A)$. So both inclusions hold and $A \cup (\Omega \setminus A) = \Omega$.

The two sets are disjoint members of $\mathcal{F}$, so we can apply the axiom on additivity from the probability function definition. If we consider $A_i = \emptyset$ for all i in the additivity axiom, this then gives $P(\emptyset) = \sum_{i=1}^{\infty} P(\emptyset)$, so $P(\emptyset) = 0$. Then if we let $A_i = \emptyset$ for $i \geq 3$, $A_1 = A$ and $A_2 = (\Omega \setminus A)$ this gives: $P(A \cup (\Omega \setminus A)) = P(A) + P(\Omega \setminus A)$ which is equal to $P(\Omega)$ since $A \cup (\Omega \setminus A) = \Omega$.

From axiom 1 of the definition of the probability function, $P(\Omega) = 1$, and rearranging the above expression gives $P(\Omega \setminus A) = 1 - P(A)$. $\square$

Property 2: if $A \subseteq B$ then $P(B) = P(A) + P(B \setminus A) \geq P(A)$

Proof. Let $A, B \in \mathcal{F}$ and $A \subseteq B$. First we need to show that $B \setminus A$ is an event. The σ -field axiom 2 only gives the complements in terms of Ω . First note that $\mathcal{F}$ is closed under intersection: for $Q, R \in \mathcal{F}$,

$$Q \cap R = \Omega \setminus ((\Omega \setminus Q) \cup (\Omega \setminus R)),$$

and the right-hand side part of the expression is built up only using axiom 2 and 3 of the σ -field definition, so $Q \cap R \in \mathcal{F}$. Now back to A and B : $B \setminus A = B \cap (\Omega \setminus A)$, since $x \in B \cap (\Omega \setminus A)$ means $x \in B \wedge x \in \Omega \wedge x \notin A$, but because B is an event $B \subseteq \Omega$ so $x \in \Omega$ is not needed. Now from axiom 2 of the σ -field definition: $\Omega \setminus A \in \mathcal{F}$, so $(B \setminus A) \in \mathcal{F}$ and $P(B \setminus A)$ is defined.

If $x \in A \cap (B \setminus A)$ then $x \in A$ and, from the definition of the complement, $x \in B \wedge x \notin A$ but $x \in A \wedge x \notin A$ cannot both be true, they contradict each other. So $A \cap (B \setminus A) = \emptyset$, A and $(B \setminus A)$ are disjoint.

If $x \in A \cup (B \setminus A)$ then $x \in A$ or $x \in B \setminus A$. If $x \in A$ then $x \in B$, since $A \subseteq B$. If $x \in B \setminus A$ then $x \in B \wedge x \notin A$, which also gives $x \in B$. Considering the converse, if $x \in B$ then either $x \in A$, or $x \notin A$ and so $x \in B \wedge x \notin A$, that is $x \in B \setminus A$. So from either case $x \in A \vee x \in B \setminus A$, which by the definition of the union is $x \in A \cup (B \setminus A)$. Both inclusions are valid, so $A \cup (B \setminus A) = B$.

A and $B \setminus A$ are disjoint members of $\mathcal{F}$, so by the finite additivity shown in Question 3.1:

$$P(A \cup (B \setminus A)) = P(A) + P(B \setminus A),$$

but we showed earlier that $A \cup (B \setminus A) = B$ so:

$$P(B) = P(A) + P(B \setminus A).$$

From the definition of the probability function P has outputs in $[0, 1]$, so $P(B \setminus A)$ is in $[0, 1]$ so it is not a negative value. Adding a non-negative value to $P(A)$ gives $P(B) \geq P(A)$ which along with the equality shown above, gives: $P(B) = P(A) + P(B \setminus A) \geq P(A)$ which completes the proof. $\square$

Property 3: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Proof. Let $A, B \in \mathcal{F}$. In Question 3.2 we showed that $\mathcal{F}$ is closed under intersection, so $(A \cap B) \in \mathcal{F}$, and we also showed that $(B \setminus A) \in \mathcal{F}$. So $P(A \cap B)$ and $P(B \setminus A)$ are both defined.

First we need to show that $A \cup B = A \cup (B \setminus A)$. If $x \in A \cup (B \setminus A)$ then $x \in A$ or $x \in (B \setminus A)$. If $x \in A$ then from the definition of the union: $x \in (A \cup B)$. If $x \in (B \setminus A)$ then $x \in B \wedge x \notin A$, so $x \in B$ and again $x \in (A \cup B)$. Considering the converse, if $x \in (A \cup B)$ then $x \in A$ or $x \in B$. If $x \in A$ we are done, but if $x \in B$ then either $x \in A$, or $x \notin A$ and so $x \in B \wedge x \notin A$, which from the definition of complement of A in B is $x \in (B \setminus A)$. So from either case $x \in A \vee x \in (B \setminus A)$, which from the definition of the union is $x \in A \cup (B \setminus A)$. Both inclusions are valid, so $A \cup B = A \cup (B \setminus A)$.

From Question 3.2, $A \cap (B \setminus A) = \emptyset$, so A and $(B \setminus A)$ are disjoint and both are members of $\mathcal{F}$. By finite additivity:

$$P(A \cup B) = P(A \cup (B \setminus A)) = P(A) + P(B \setminus A).$$

Then we need to show $B = (A \cap B) \cup (B \setminus A)$. If $x \in (A \cap B) \cup (B \setminus A)$ then $x \in (A \cap B)$ or $x \in (B \setminus A)$. From the definition of the intersection, $x \in (A \cap B)$ so, $x \in A \wedge x \in B$, so $x \in B$. From the definition of the complement: $x \in B \setminus A$ means $x \in B \wedge x \notin A$, so again $x \in B$. Considering the converse, if $x \in B$ then either $x \in A$, and so $x \in A \wedge x \in B$, that is $x \in (A \cap B)$; or $x \notin A$, and so $x \in B \wedge x \notin A$, that is $x \in (B \setminus A)$. So from each of the cases: $x \in (A \cap B) \vee x \in (B \setminus A)$, and from the definition of the union: $x \in (A \cap B) \cup (B \setminus A)$. Both inclusions hold, so $B = (A \cap B) \cup (B \setminus A)$.

If $x \in (A \cap B) \cap (B \setminus A)$ then $x \in A \wedge x \in B$ and $x \in B \wedge x \notin A$, but that is a contradiction. So $(A \cap B) \cap (B \setminus A) = \emptyset$ and by finite additivity:

$$P(B) = P(A \cap B) + P(B \setminus A).$$

This can be rearranged to give: $P(B \setminus A) = P(B) - P(A \cap B)$. If we then plug this into the expression for $P(A \cup B)$ we get:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

which completes the proof. $\square$